

Marvin Gülhan



EDUCATION

University of Bonn

Master of Science in Computer Science

Bonn, Germany

Oct. 2025 – present

University of Bonn

Bachelor of Science in Computer Science (Grade: 1.2)

Bonn, Germany

Oct. 2022 – Sep. 2025

Rhein-Sieg-Gymnasium

Abitur (Grade: 1.0)

Sankt Augustin, Germany

Aug. 2014 – June 2022

German International School of Silicon Valley

Semester Abroad

Mountain View (CA), USA

Aug. 2019 – Jan. 2020

WORK EXPERIENCE & INTERNSHIPS

Research Assistant

Fraunhofer Institute FKIE

Wachtberg, Germany

Nov. 2023 – present

- Algorithm engineering for online graph problems
- Research on RL algorithms for dynamic planning problems

Formula Student Team Member

Bonn-Rhein-Sieg University of Applied Sciences (HBRS)

Sankt Augustin, Germany

Sep. 2023 – Jun. 2025

- Member of the Formula Student team at HBRS
- CFD simulation and aerodynamic design of the rear wing for the 2024 race car

Student Internship

German Aerospace Center (DLR)

Cologne, Germany

Jan. 2020

- Insights into aerospace engineering and research methods
- Use of clustering methods for anomaly detection in fuel combustion of hybrid engines

PUBLICATIONS & PROJECTS

Airlift Challenge: A Competition for Optimizing Cargo Delivery

2025

- Co-author of a research paper on optimizing cargo delivery for Airlift Operations
- Developed a routing algorithm coordinating heterogeneous aircraft across subgraphs to manage bottlenecks and disruptions in dynamic conditions, achieving 2nd place out of 40 teams

Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems

2026

- Co-author of a peer-reviewed paper accepted at the 2026 International Conference on Military Communications and Information Systems (ICMCIS), Bath, UK
- Demonstrated that Graph Reinforcement Learning policies can effectively solve complex, dynamic planning problems such as the Airlift Planning Problem (in-press)

HONORS

Deutschlandstipendium

Federal Ministry of Education and Research

Bonn, Germany

Oct. 2024 & Oct. 2025

- Merit-based scholarship for outstanding academic performance

CORE SKILLS

Programming Languages: Python, C++, Java

Technical Skills: Graph RL, Multi-Agent RL, GNNs, PyTorch, HPC, CFD simulations

Languages: German (C2), English (C1), French (B1), Spanish (B1)